

Customer Support Ticket Analyzer

Objective

To analyze customer support tickets using Python and derive meaningful insights related to:

- Ticket priority distribution
- Common issue patterns (keyword analysis)
- Customer complaint trends
- Service quality indicators
- Text-based insights from issue descriptions

Database Description

This project simulates a **customer support system** where tickets are raised by users.

It captures:

- Customer details
- Issue descriptions
- Ticket priority levels

Tables and Structure

ticket_data

Column Name	Description
Ticket_No	Unique ID for each ticket
Customer_Name	Name of the customer
Issue_Description	Description of the problem
Priority	Ticket priority (High / Medium / Low)

Tools and Technologies

- **Python** → Core programming
- **Built-in Functions** → split(), join(), lower(), replace()
- **Loops & Conditions** → Data processing
- **Functions** → Reusable logic

Analysis:

Initial Ticket Data

Final Cleaned Ticket Data:

```
Ticket 1 | Customer: Ravi | Issue: internet not working | Priority: High
Ticket 2 | Customer: Meera | Issue: slow response very poor service | Priority: Low
Ticket 3 | Customer: Sam | Issue: great support issue resolved | Priority: High
Ticket 4 | Customer: Anu | Issue: okayayayayayayay need help | Priority: Medium
Ticket 5 | Customer: Rakesh | Issue: not bad but slow | Priority: Low
Ticket 6 | Customer: Divya | Issue: excellent guidance very helpful | Priority: High
Ticket 7 | Customer: Arjun | Issue: good support and good behaviour | Priority: Medium
Ticket 8 | Customer: Kiran | Issue: poor handling of technical issue | Priority: High
Ticket 9 | Customer: Leela | Issue: satisfied could be better | Priority: Low
Ticket 10 | Customer: Nisha | Issue: good service quick response | Priority: Medium
Ticket 11 | Customer: John | Issue: internet connection is very slow | Priority: High
Ticket 12 | Customer: Priya | Issue: login page not opening properly | Priority: Medium
Ticket 13 | Customer: Arun | Issue: poor network signal in my area | Priority: High
Ticket 14 | Customer: Kavya | Issue: good support but delayed response | Priority: Low
Ticket 15 | Customer: Manoj | Issue: excellent help from support team | Priority: Medium
Ticket 16 | Customer: Sneha | Issue: internet disconnects frequently | Priority: High
Ticket 17 | Customer: Rahul | Issue: poor service quality and slow speed | Priority: High
Ticket 18 | Customer: Asha | Issue: billing issue not resolved | Priority: Medium
Ticket 19 | Customer: Vignesh | Issue: good response and quick resolution | Priority: Low
Ticket 20 | Customer: Deepa | Issue: excellent customer service experience | Priority: Medium
```

Priority Analysis

Priority Analysis:

High Priority Tickets: 8

Medium Priority Tickets: 7

Low Priority Tickets: 5

Longest Issue Description

Ticket With Longest Issue Description:

Ticket Number: 17

Customer Name: Rahul

Issue: poor service quality and slow speed

Word Count: 6

Unique Words

Unique Words Used:

Count: 49

Words: ['and', 'arun', 'bad', 'be', 'behaviour', 'better', 'billing', 'but', 'connection', 'could', 'customer', 'd

Words: ['and', 'arun', 'bad', 'be', 'behaviour', 'better', 'billing', 'but', 'connection', 'could',
'customer', 'delayed', 'disconnects', 'excellent', 'experience', 'frequently', 'from', 'good', 'great',
'guidance', 'handling', 'help', 'helpful', 'internet', 'is', 'issue', 'login', 'need', 'not', 'of', 'okayayay',
'opening', 'page', 'poor', 'properly', 'quality', 'quick', 'resolution', 'resolved', 'response',
'satisfied', 'service', 'slow', 'speed', 'support', 'team', 'technical', 'very', 'working']

Key Insights

- High-priority tickets indicate critical service issues
- Frequent keywords like "slow", "poor" highlight service gaps
- Positive words like "good", "excellent" indicate satisfaction
- Some tickets contain longer descriptions → more complex issues
- Repeated words show common problem trends

Conclusion

- Focus on resolving high-priority tickets quickly
- Improve areas where negative keywords appear frequently
- Use positive feedback to identify strong service areas
- Analyze repeated issue patterns for system improvements
- Enhance support quality using text-based insights

Filename: Project Documentation - Python
Directory: C:\Users\LAVAMADHAN\OneDrive\Documents
Template: Normal.dotm
Title:
Subject:
Author: Madhan Raj T
Keywords:
Comments:
Creation Date: 02-05-2026 17:48:00
Change Number: 27
Last Saved On: 02-05-2026 18:28:00
Last Saved By: Madhan Raj T
Total Editing Time: 38 Minutes
Last Printed On: 02-05-2026 18:29:00
As of Last Complete Printing
 Number of Pages: 3
 Number of Words: 308 (approx.)
 Number of Characters: 1,757 (approx.)